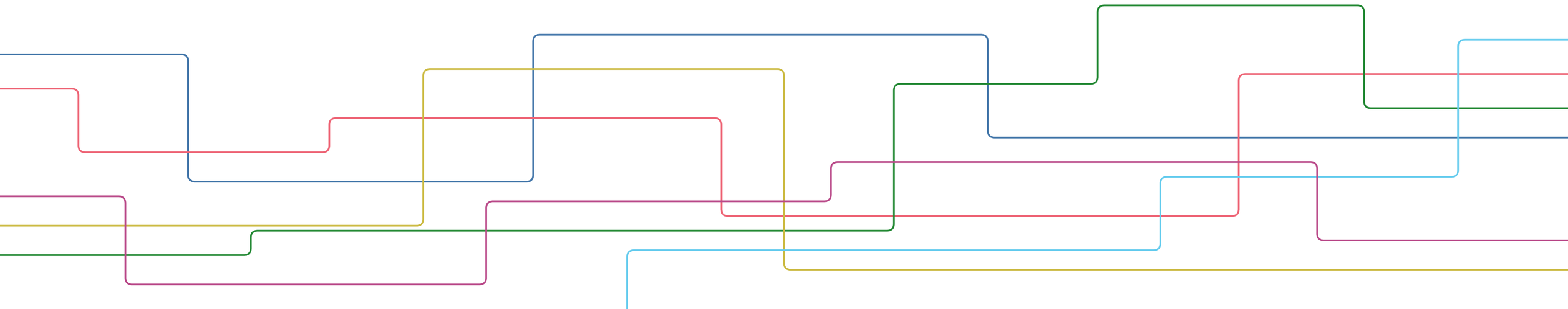
Programming for Data Science

– Gaussian Processes

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School of Electrical Engineering and Computer Science
KTH Royal Institute of Technology
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Outline

Gaussian distribution

- Covariance
- Sampling random vectors
- Conditional Gaussian distributions

Gaussian process

- Fundamentals and computation
- Gaussian process regression
- Kernel functions

Gaussian distribution

- A random vector $\mathbf{x} = [x_1, x_2, \ldots, x_D]^T$ is said to have a multivariate Gaussian distribution if all linear combinations of the vector $\mathbf{x}$ are Gaussian-distributed:

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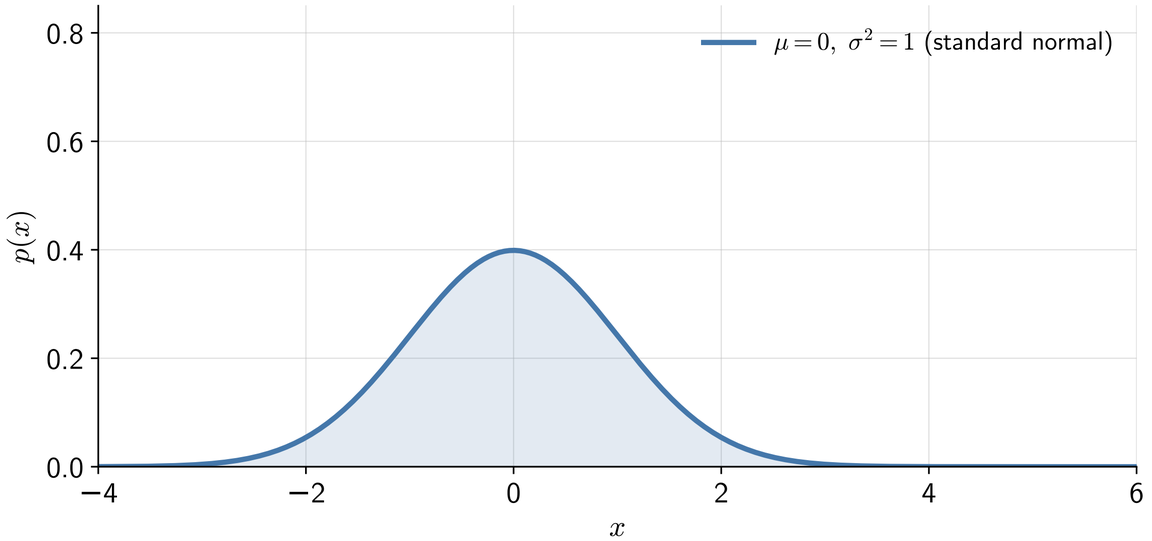
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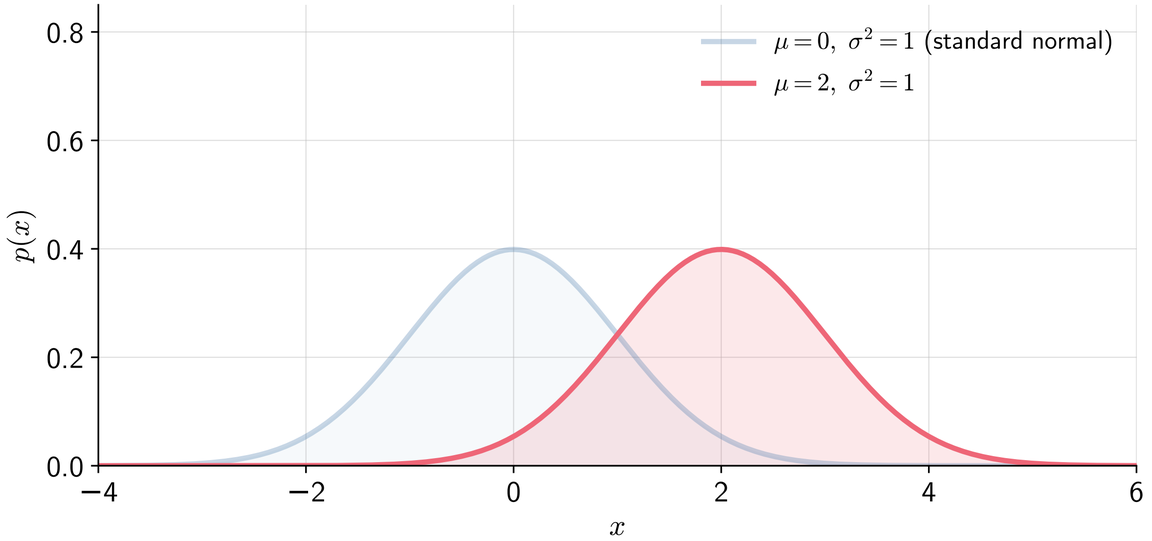


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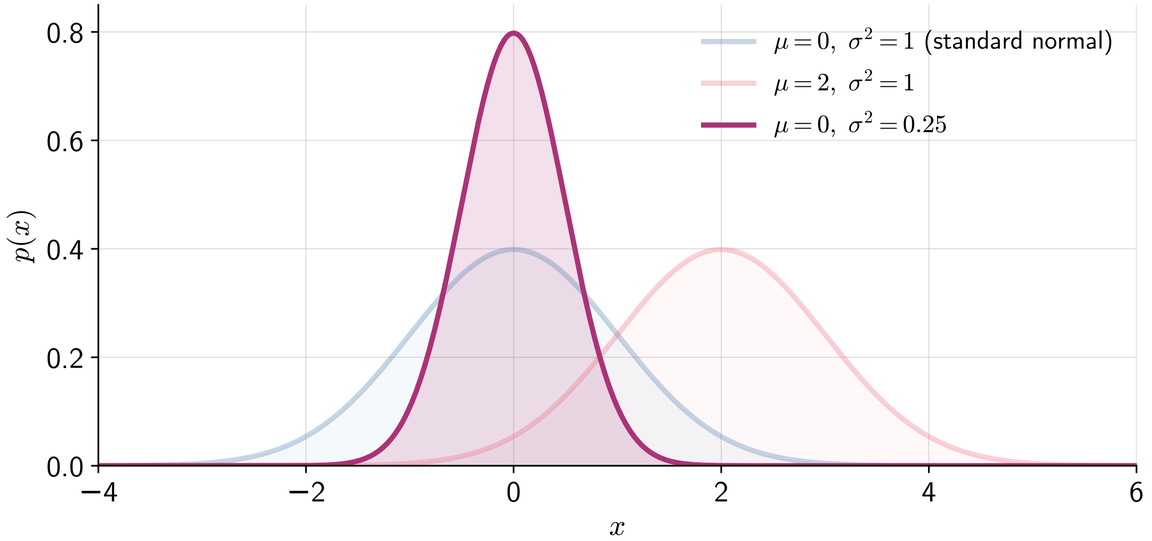


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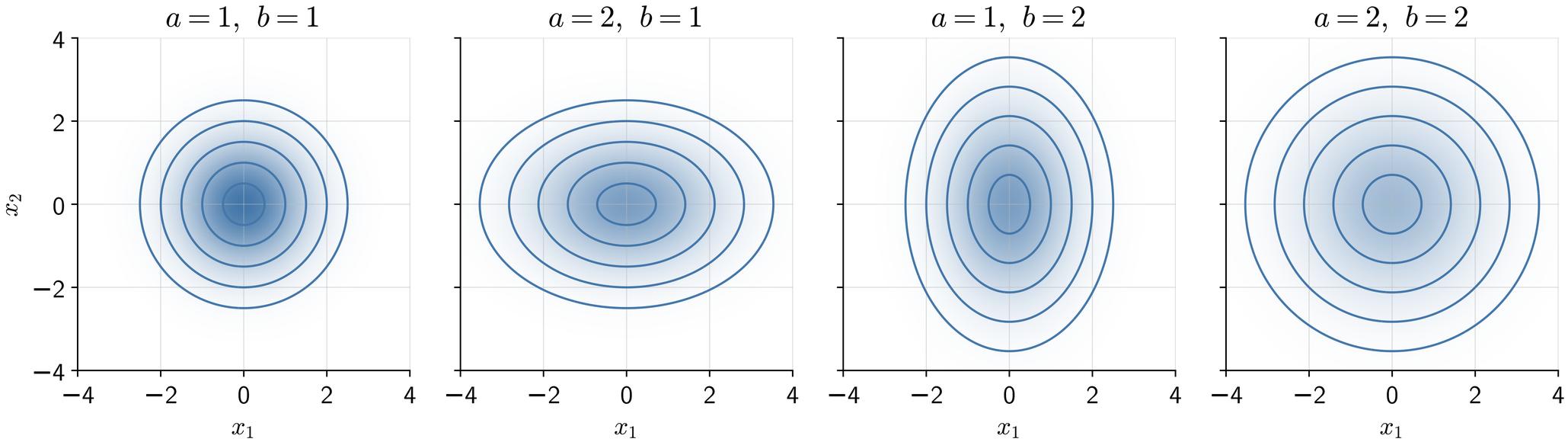
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Interpretation of the covariance matrix

- The diagonal of the covariance matrix controls the scaling/marginal variances

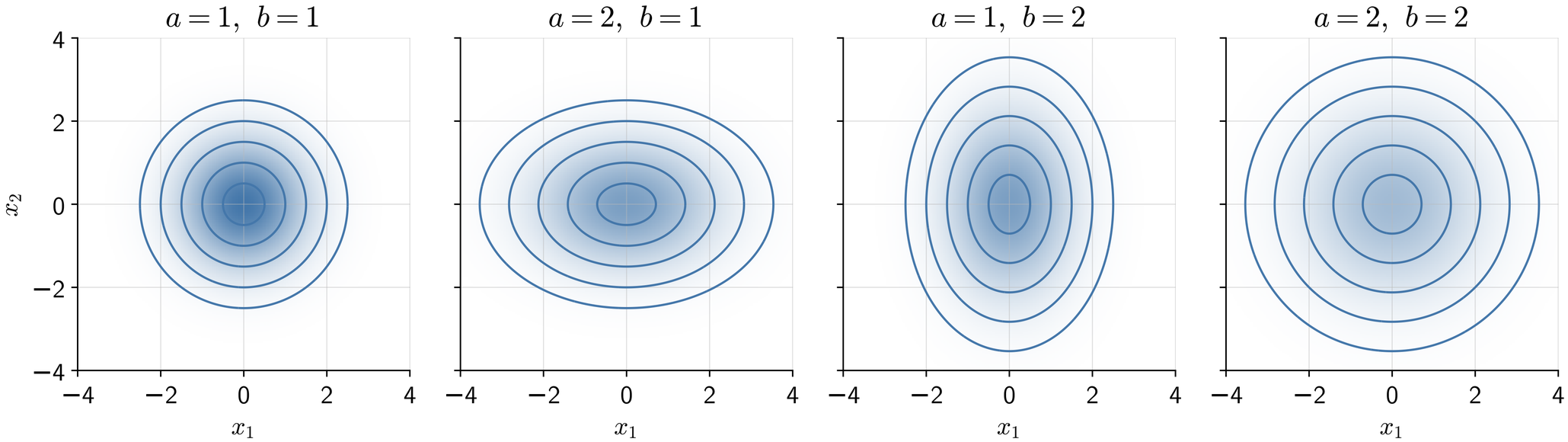
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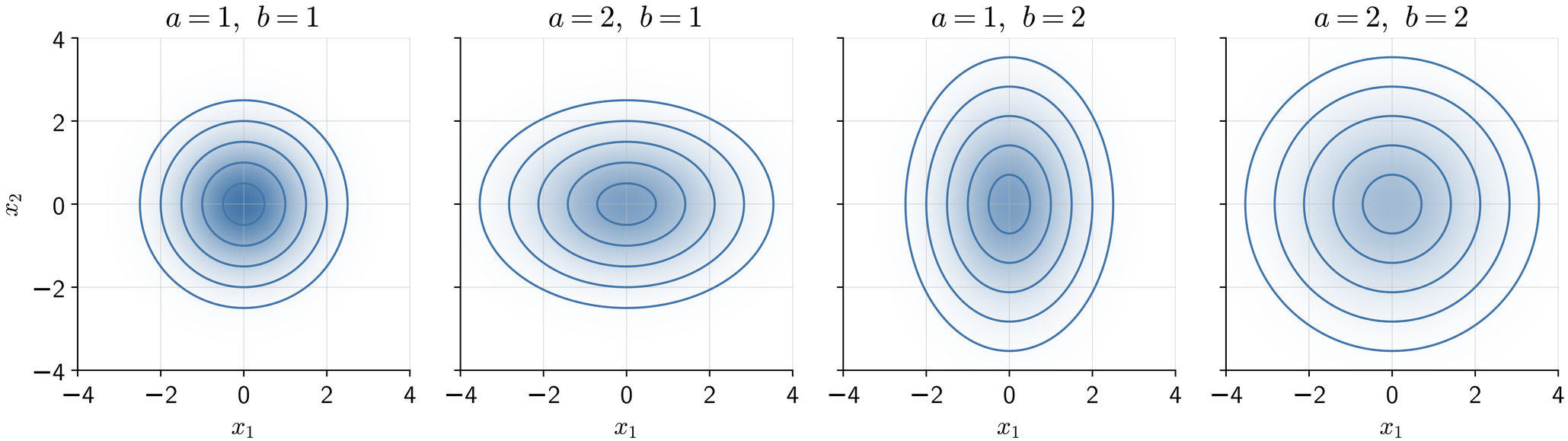


- If Σ is diagonal, then x_1 and x_2 are uncorrelated? True or false?

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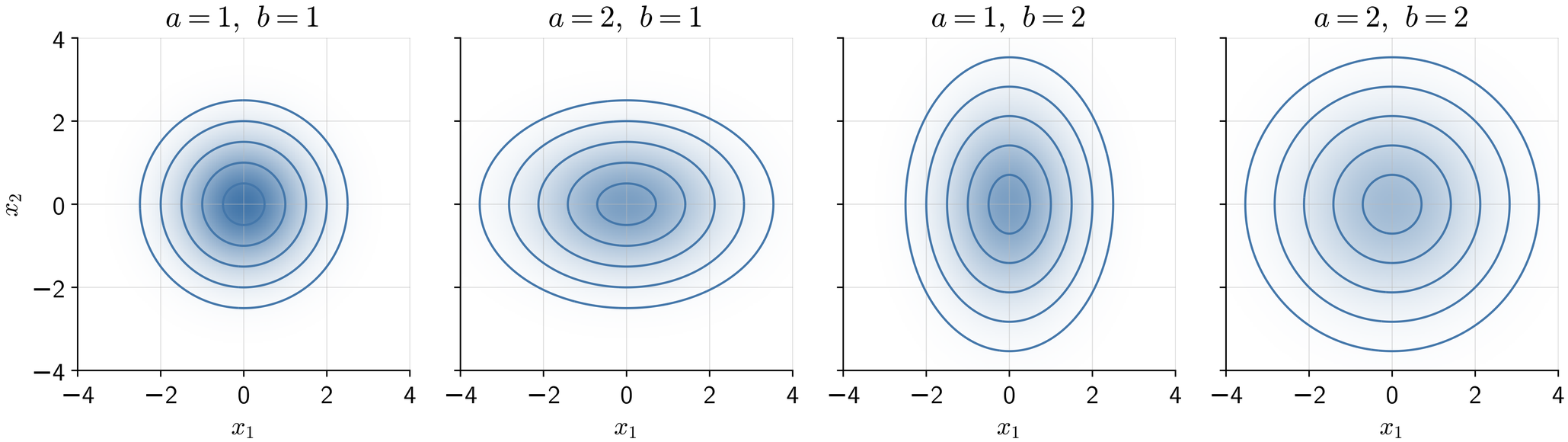


- What is the volume (integral) of the density?

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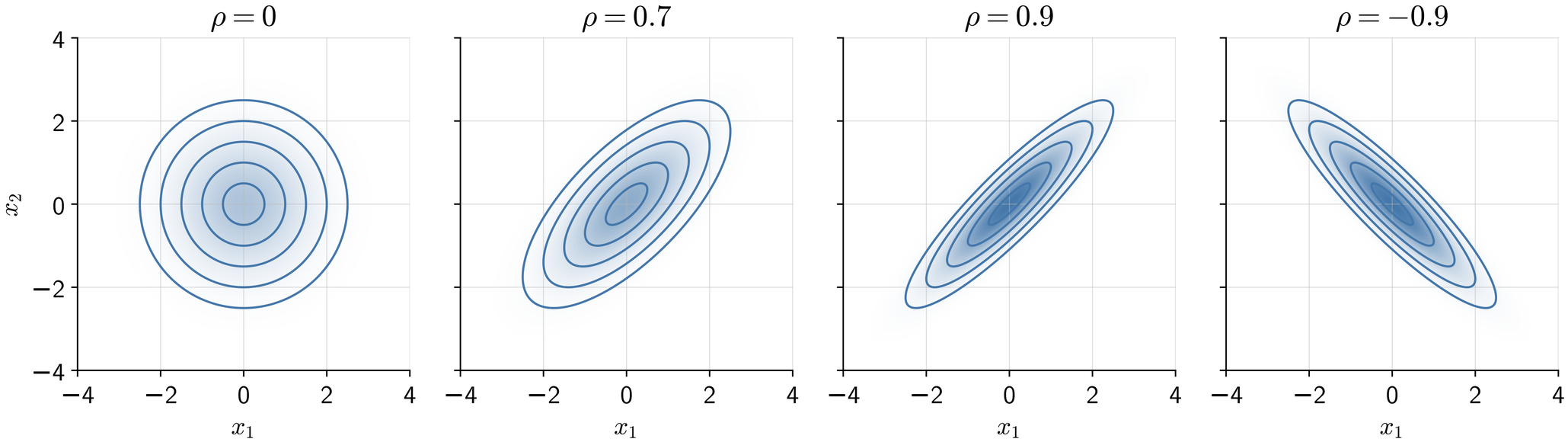


- Which of the four densities has the highest peak?

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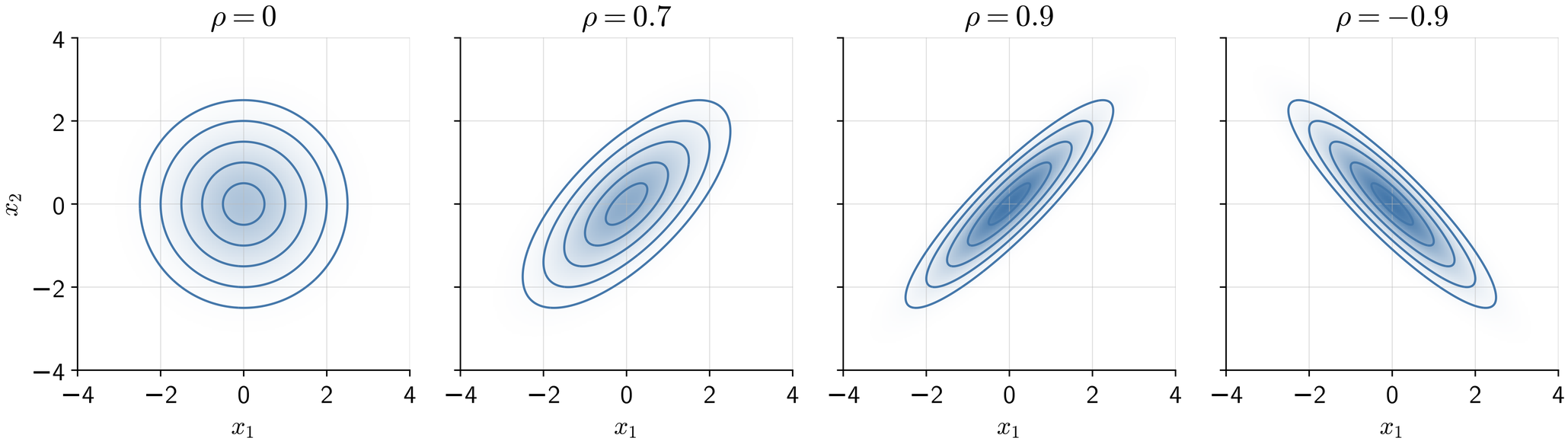
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Inverse of whitening operation

Sampling from a multivariate Gaussian

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Gaussian samples

```python
import numpy as np

x = np.random.randn(100, 2)     # sample 100 observations (D = 2)

m = [-2.0, 1.0]                 # mean vector of y
V = [[1.0, 0.7], [0.7, 2.0]]    # covariance matrix of y

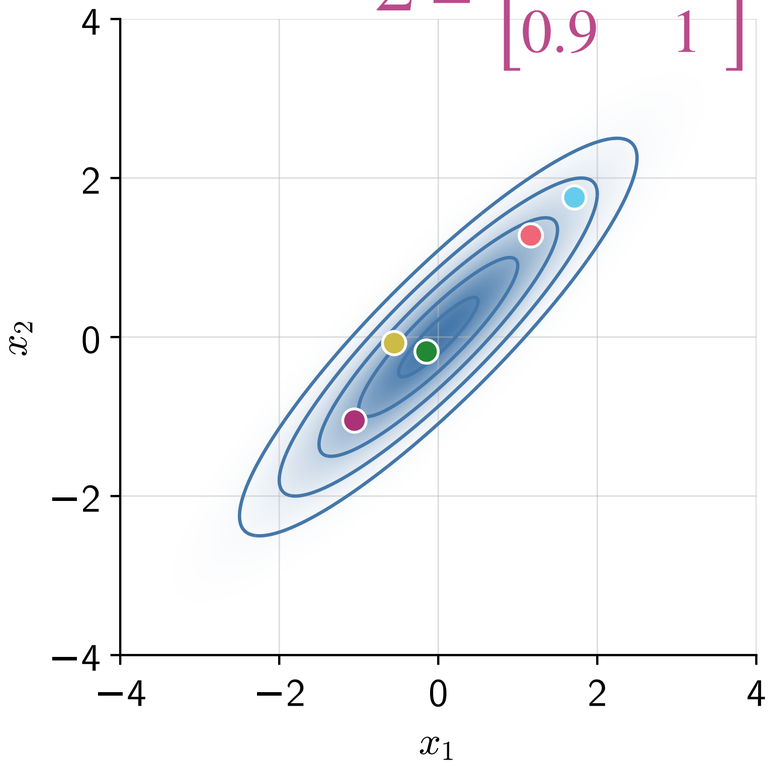
L = np.linalg.cholesky(V)       # matrix square root (Cholesky factor)
y = (x @ L.T) + m               # transform x samples into y

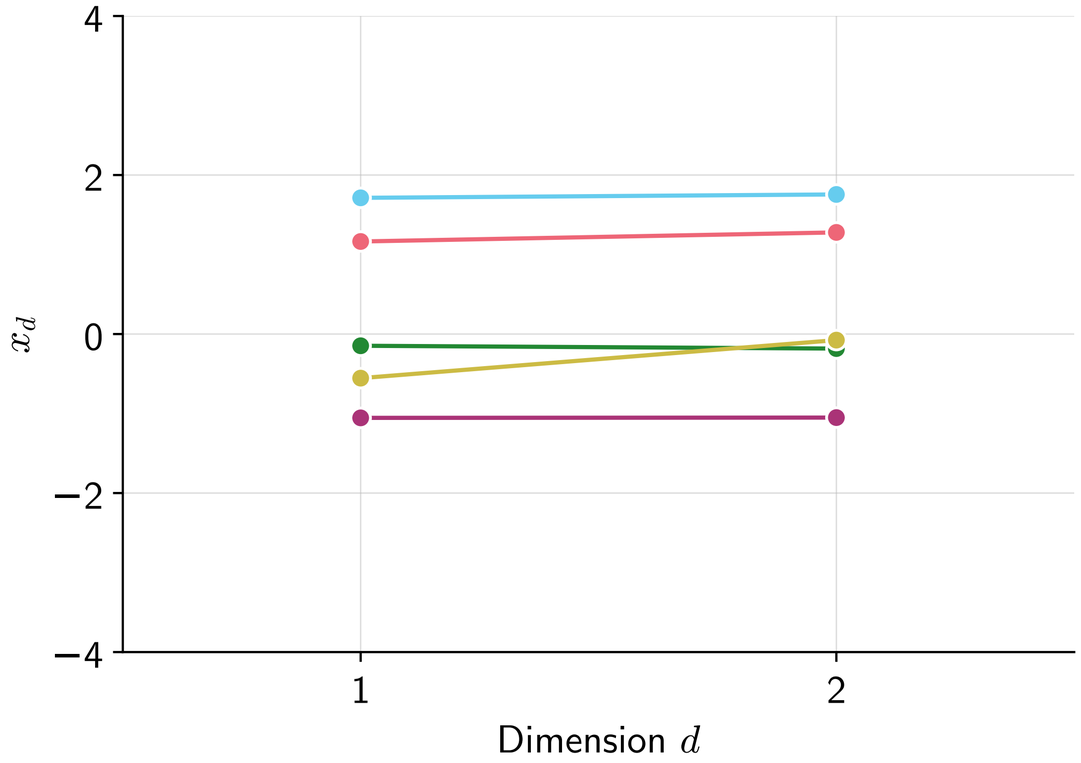
```

Samples in higher dimensions

- Samples in $D = 2$

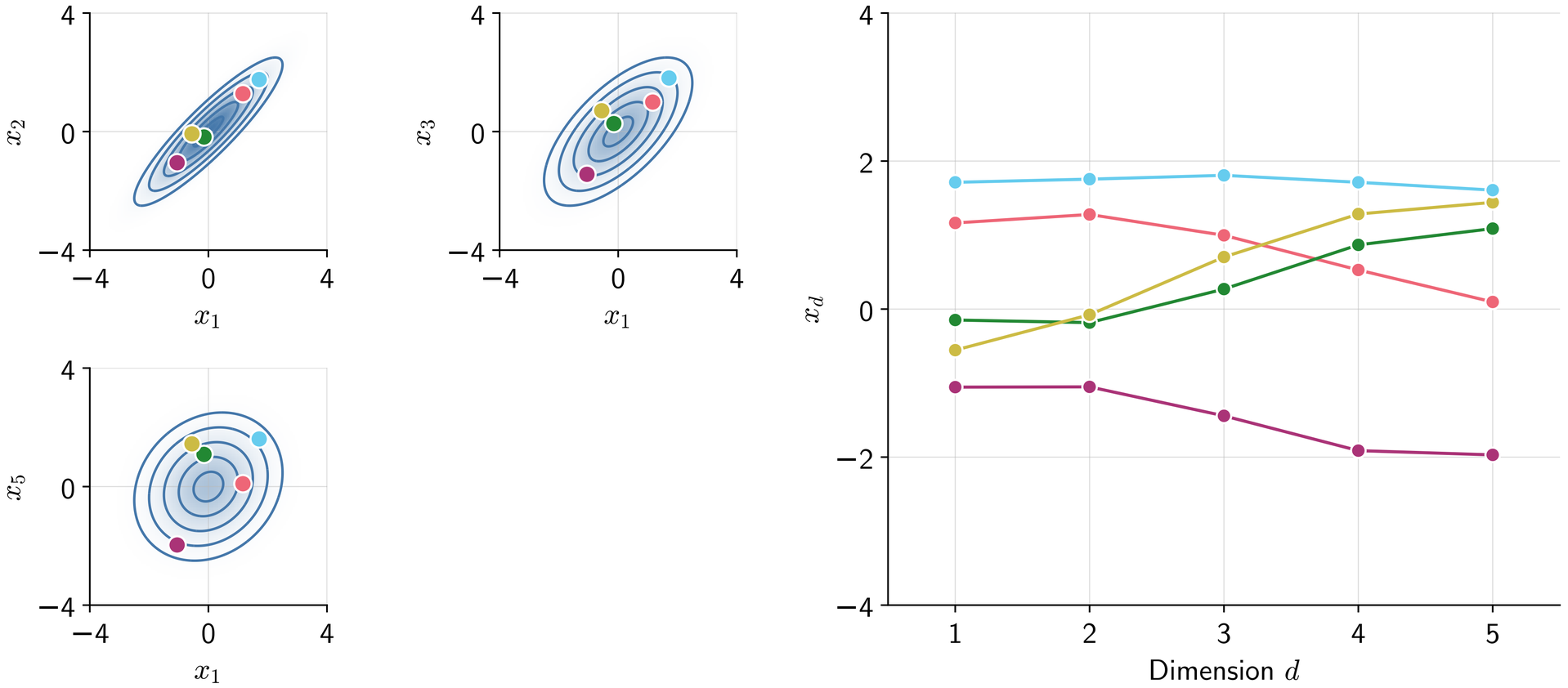
$$\Sigma = \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}$$





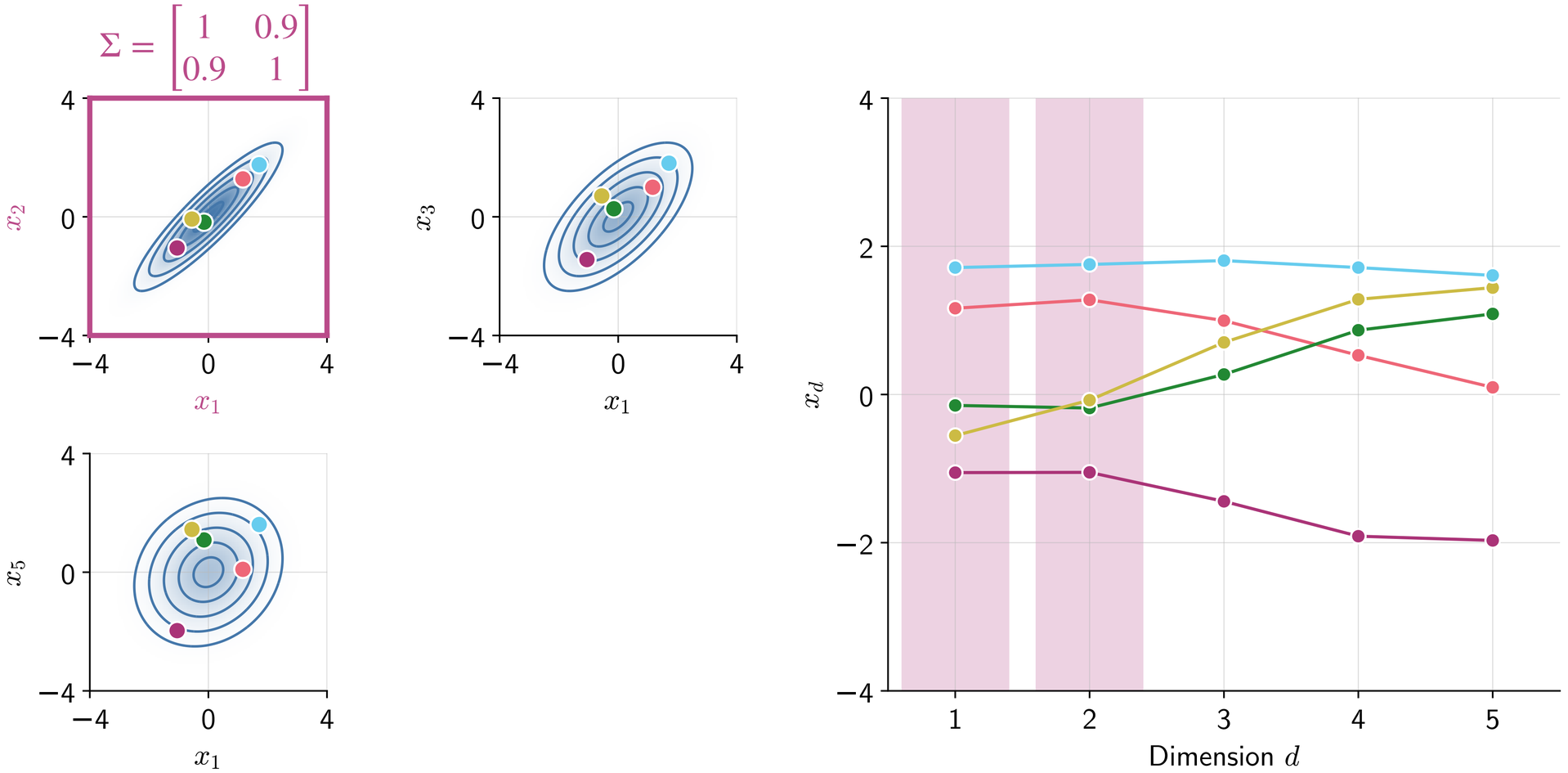
Samples in higher dimensions

- Samples in $D = 5$



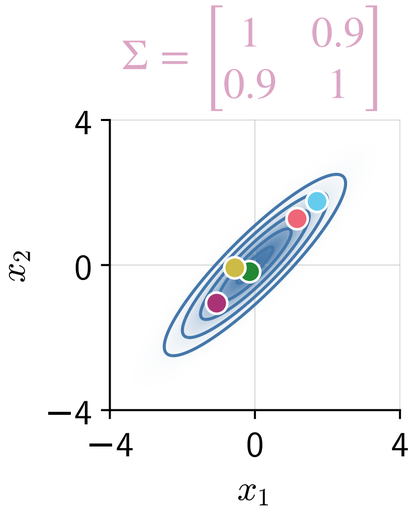
Samples in higher dimensions

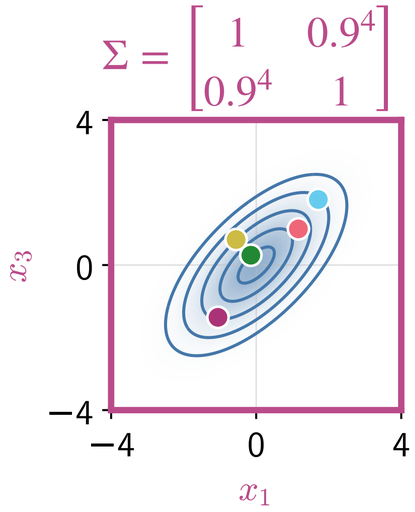
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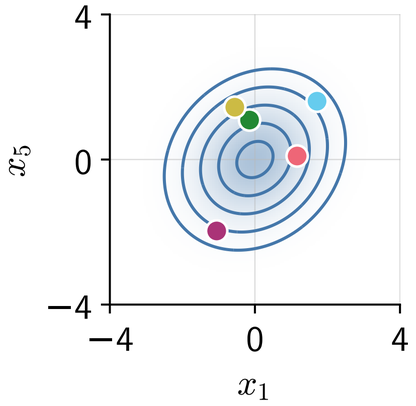


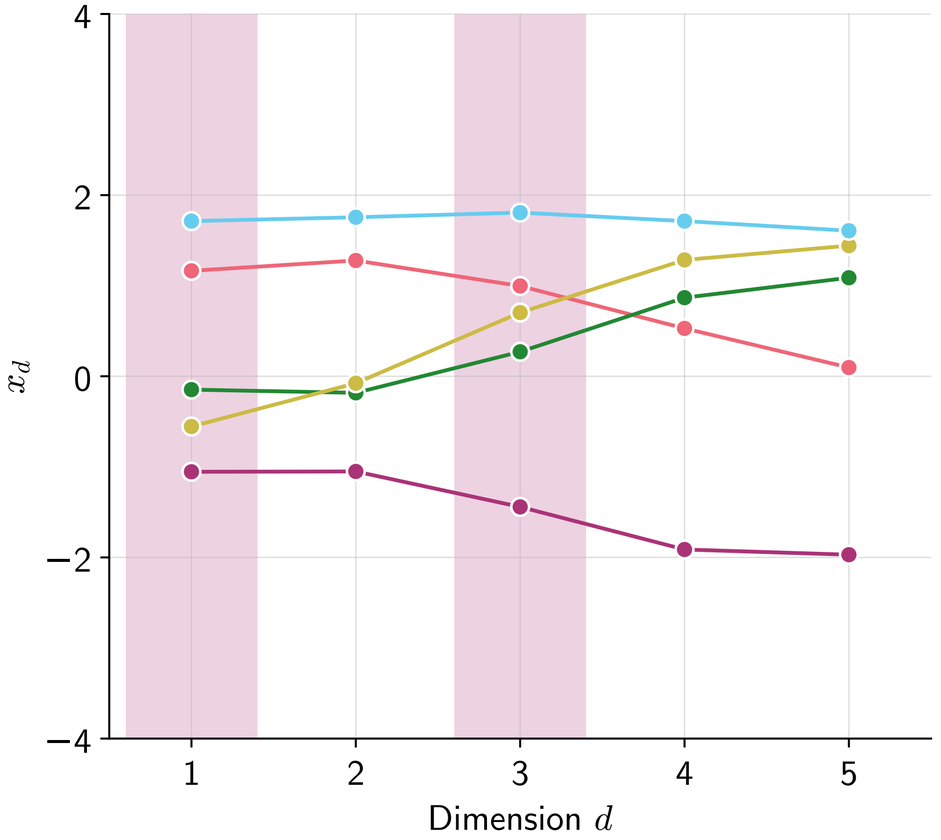
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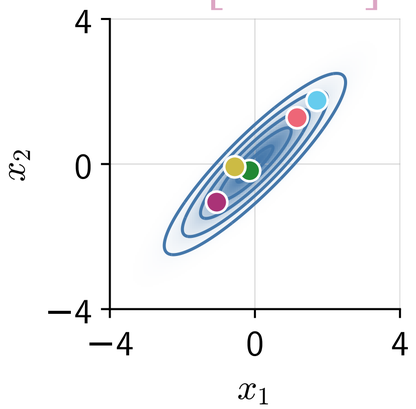




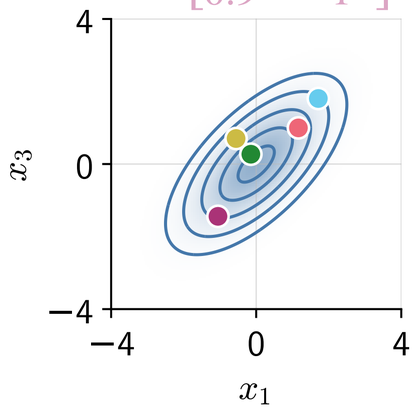
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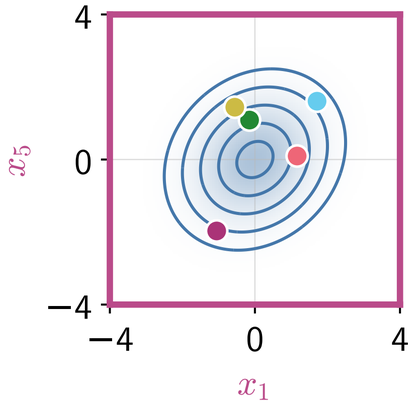
- Samples in $D = 5$

$$\Sigma = \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}$$

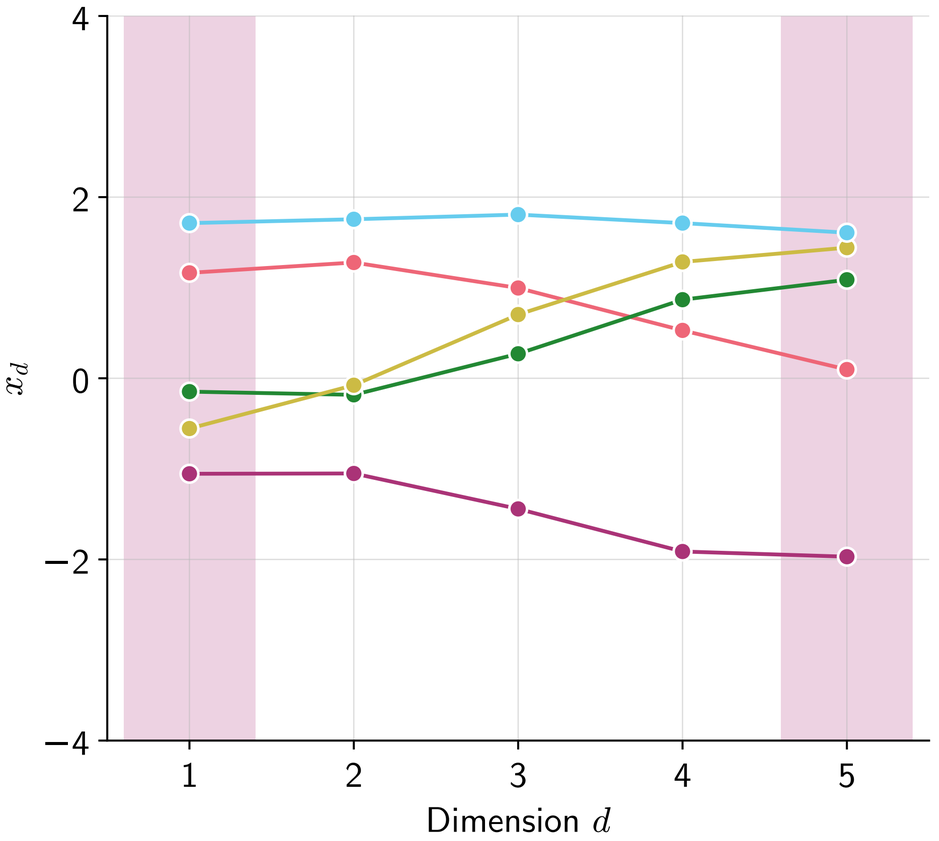


$$\Sigma = \begin{bmatrix} 1 & 0.9^4 \\ 0.9^4 & 1 \end{bmatrix}$$



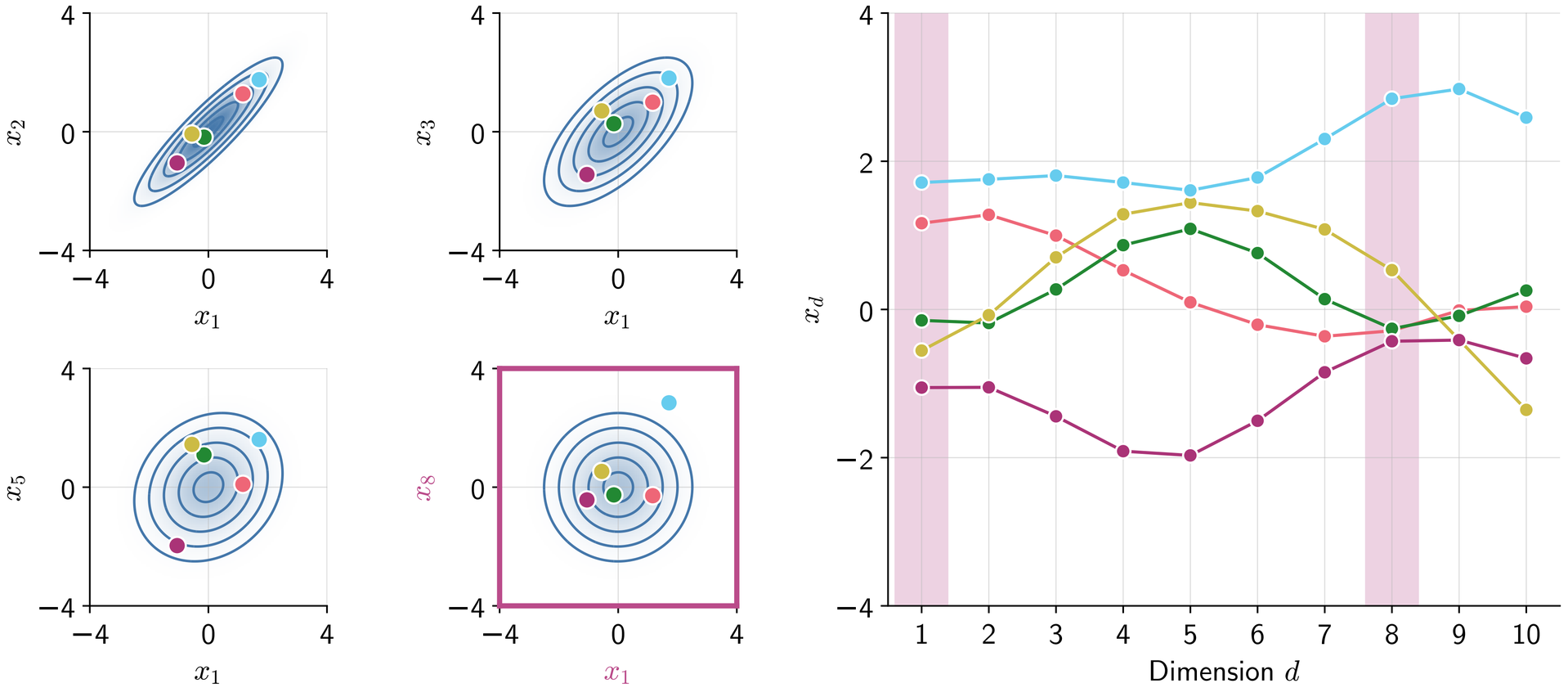


$$\Sigma = \begin{bmatrix} 1 & 0.9^{16} \\ 0.9^{16} & 1 \end{bmatrix}$$



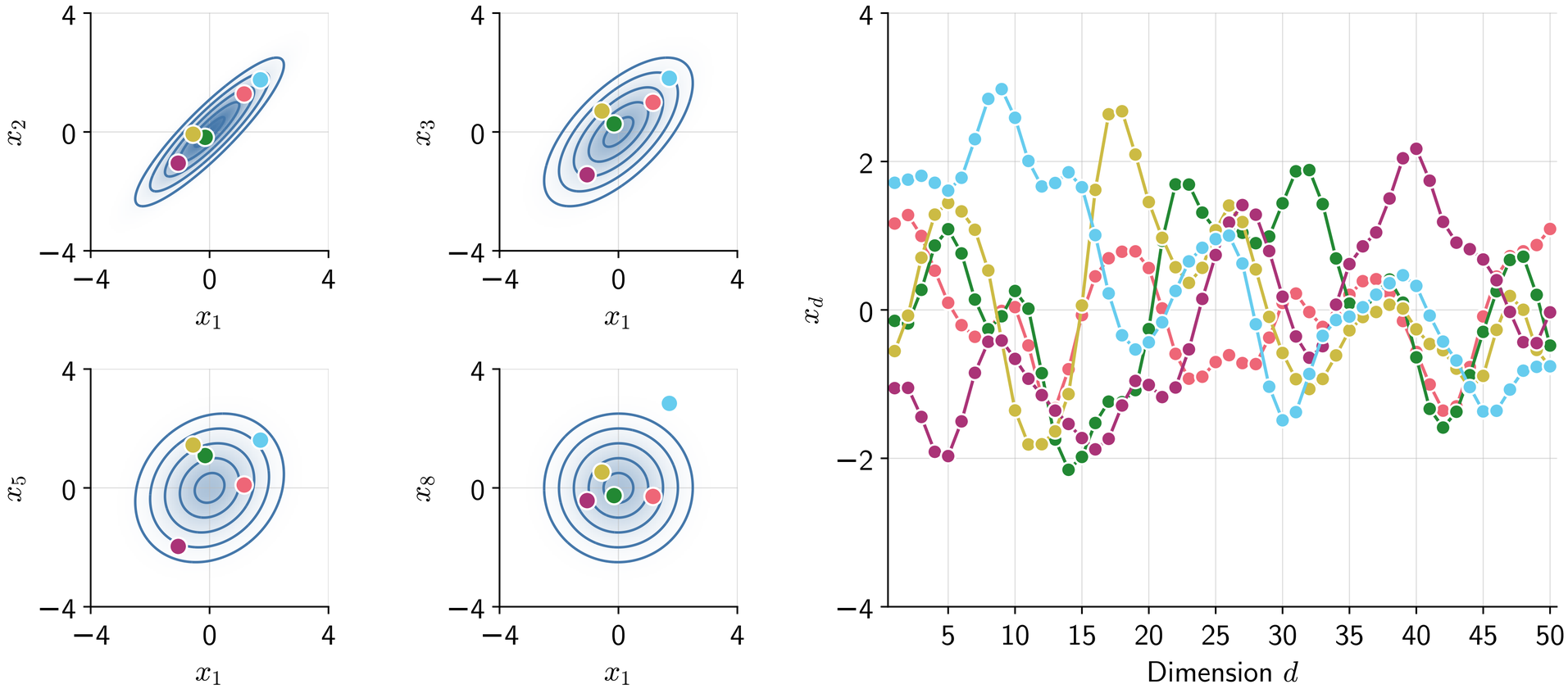
Samples in higher dimensions

- Samples in $D = 10$



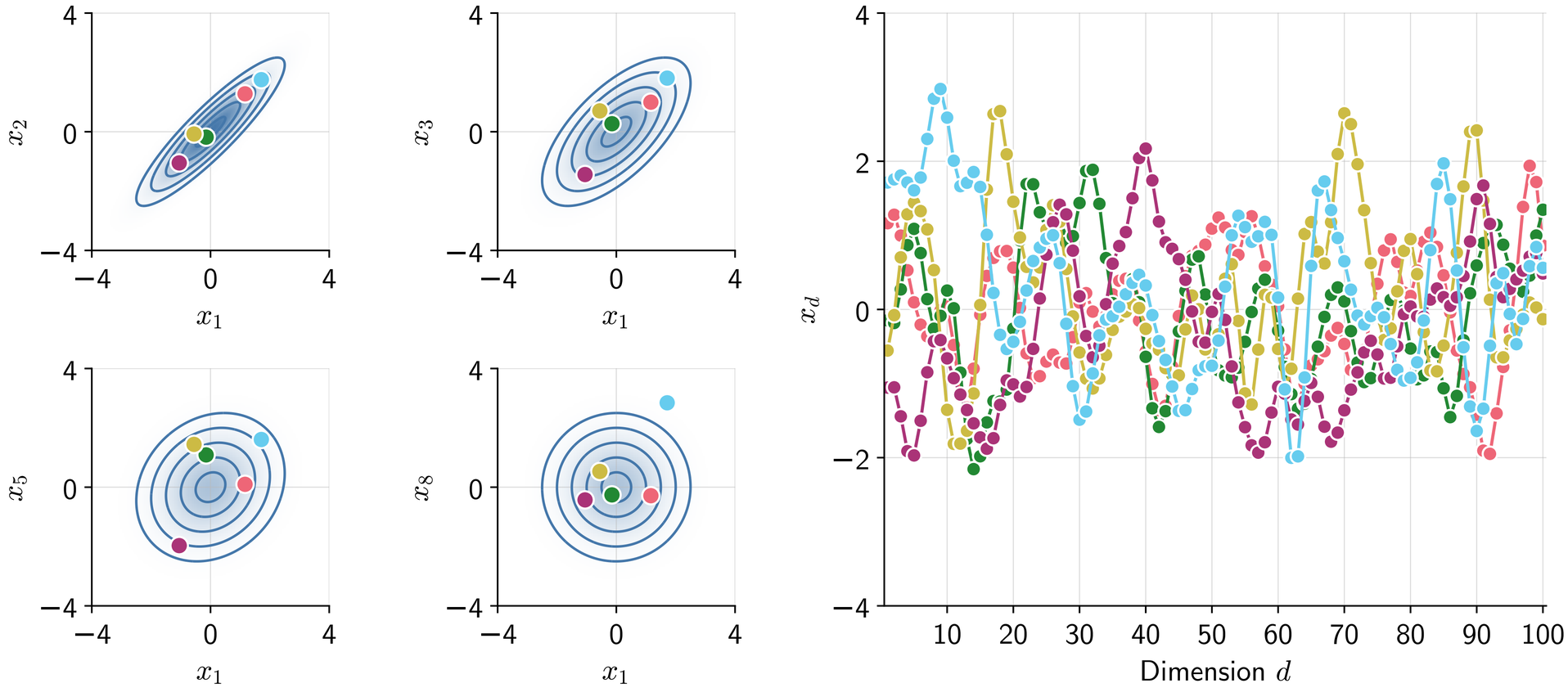
Samples in higher dimensions

- Samples in $D = 50$



Samples in higher dimensions

- Samples in $D = 100$



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- *We say, $p(y \mid x)$ is the conditional probability of y given (conditioned on) x*

Conditioning a multivariate Gaussian

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- Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be jointly Gaussian distributed:

$$p(\mathbf{x}_1, \mathbf{x}_2) = \mathcal{N}\left(\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} \mid \begin{bmatrix} \mathbf{m}_1 \\ \mathbf{m}_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \right)$$

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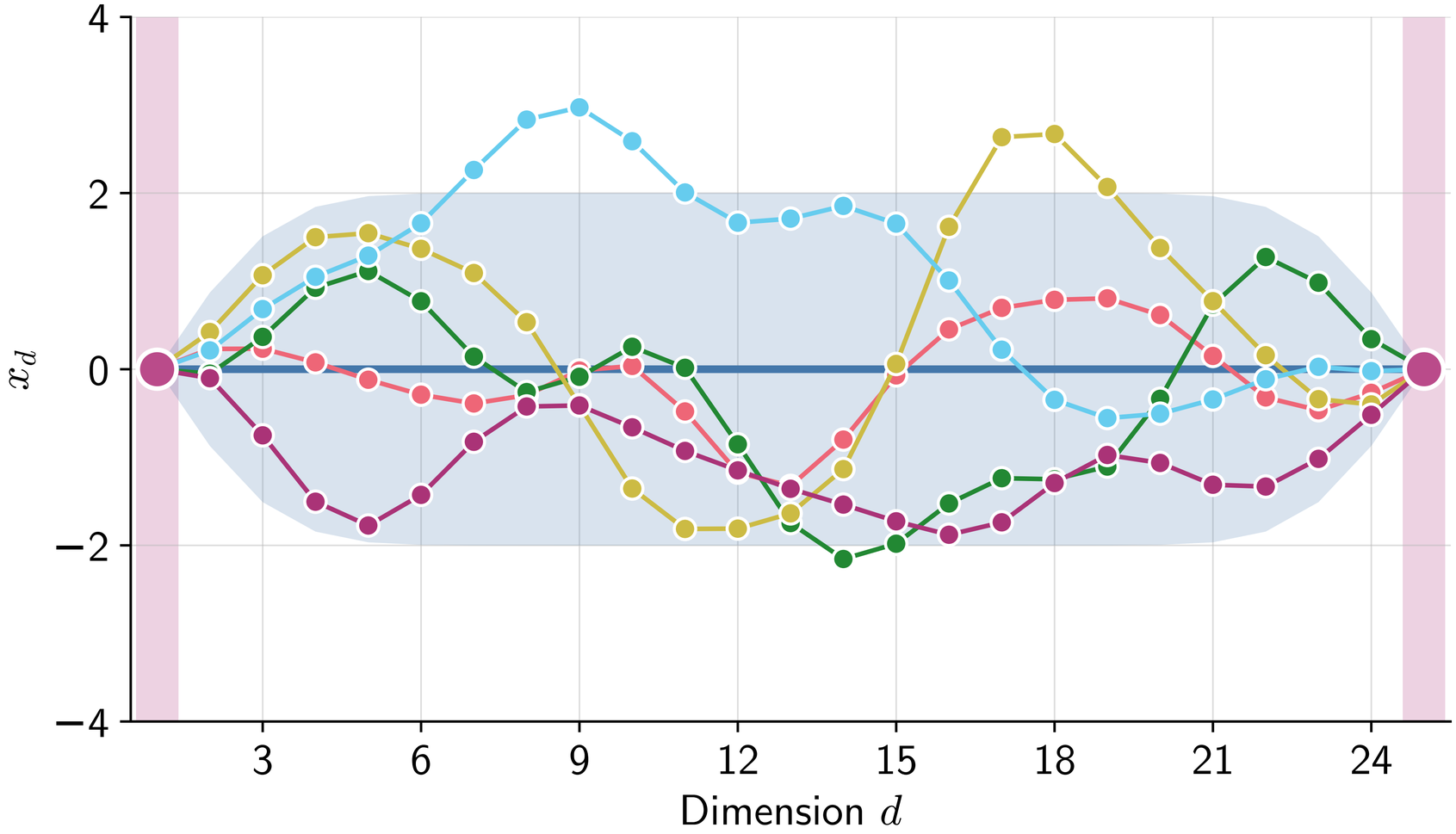
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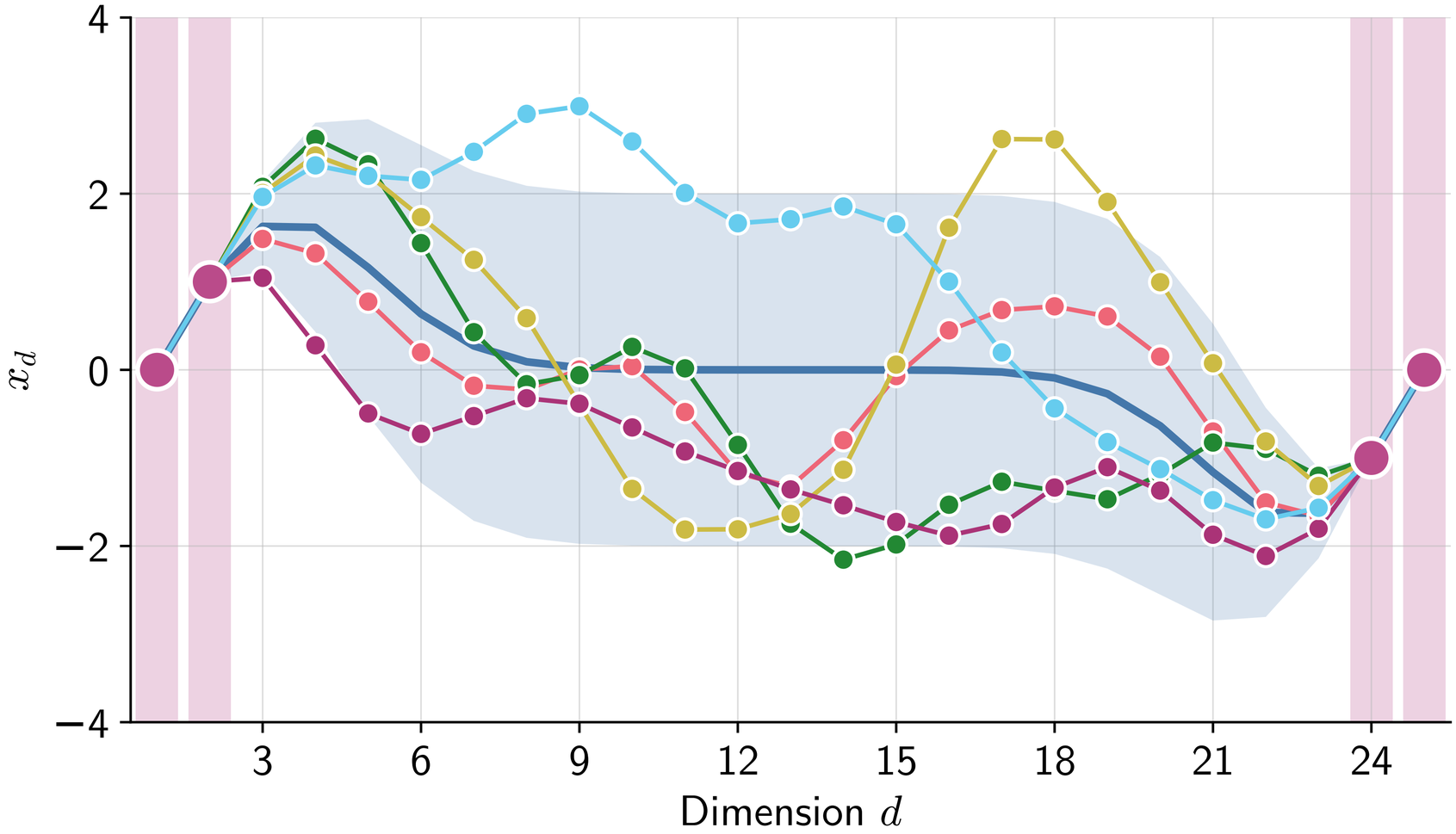
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mean vector covariance matrix

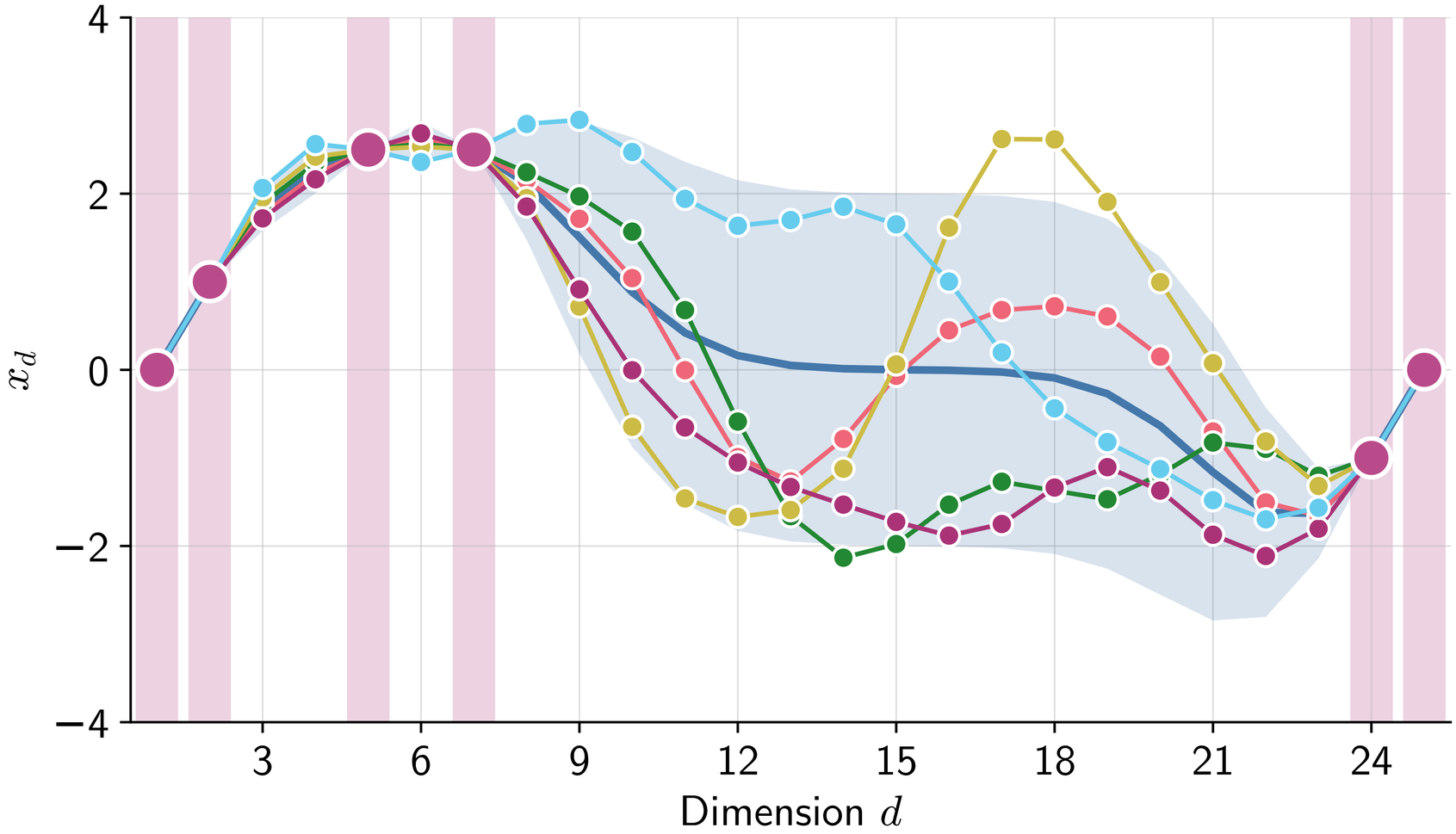
Conditioning a multivariate Gaussian



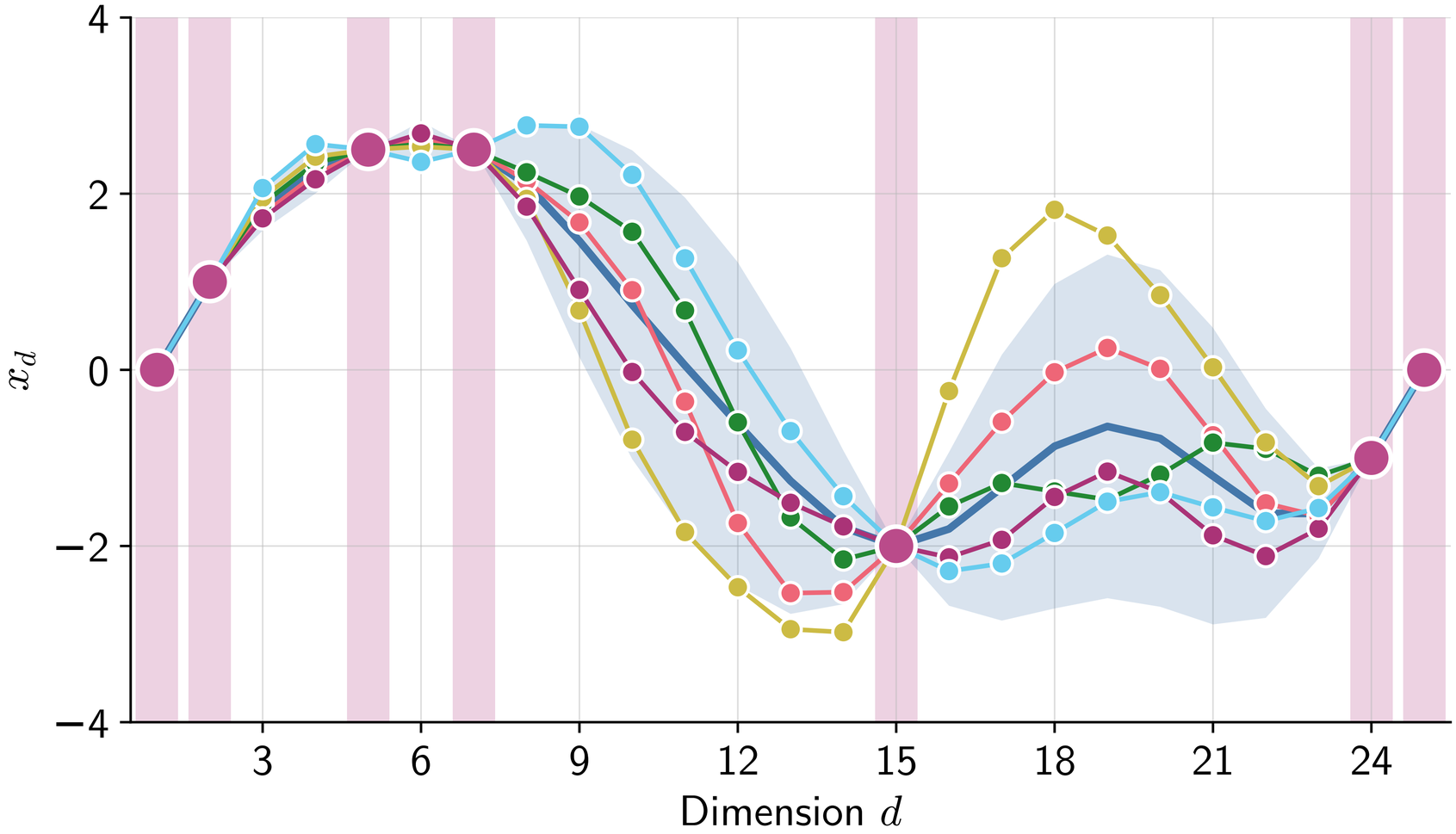
Conditioning a multivariate Gaussian



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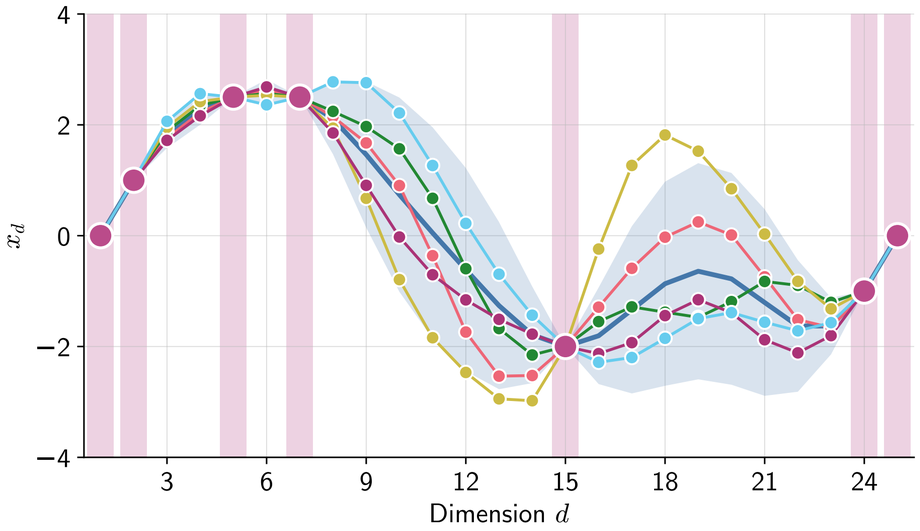


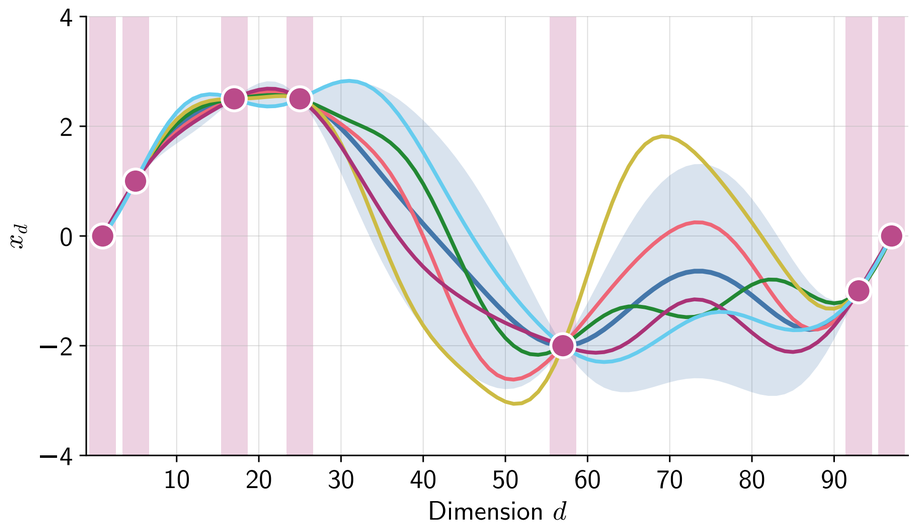
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- Let's now consider the case of $D = 100$

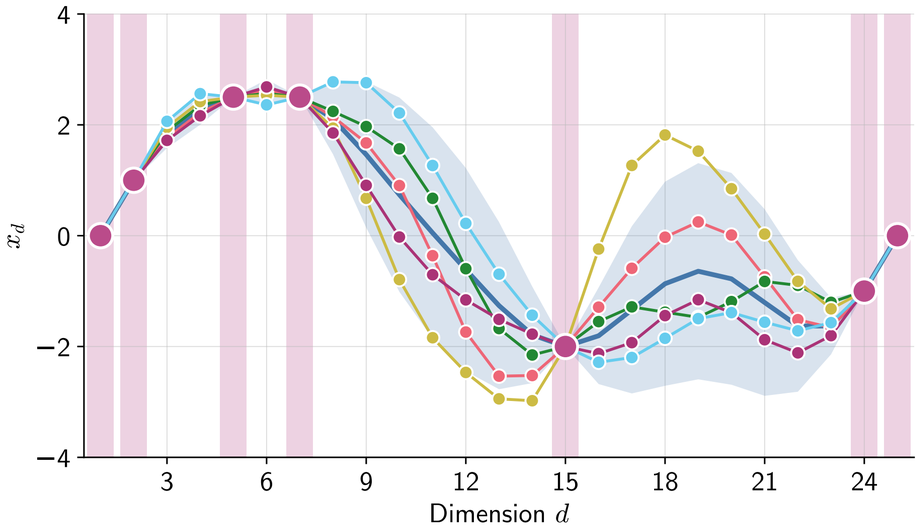


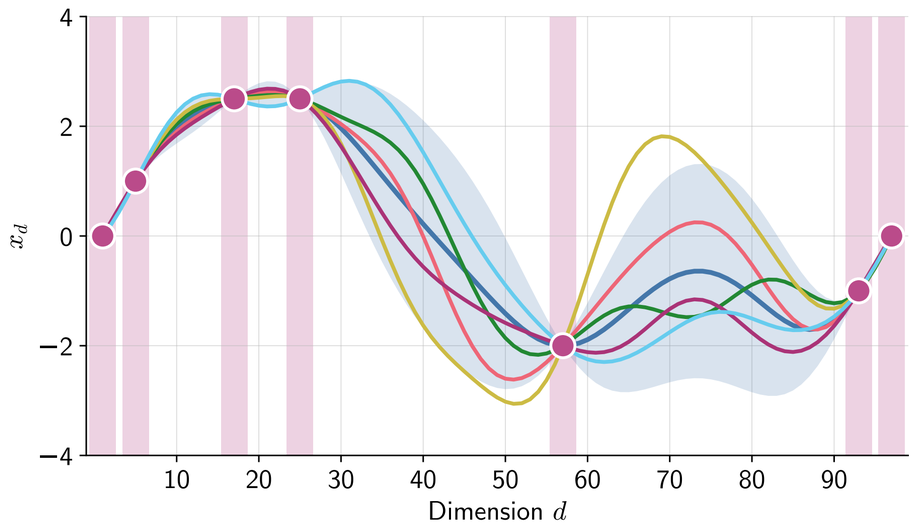


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- By computing conditional Gaussian distributions, we can do non-linear regression!

Gaussian process

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- It is fully specified by its mean function $m(\,\cdot\,)$ and its covariance function $k(\,\cdot\,,\cdot\,)$

$$\mathbb{E}_{f\sim\mathrm{GP}}\left[f(\mathbf{x})\right] = m(\mathbf{x}) \qquad \mathbf{cov}\left[f(\mathbf{x}), f(\mathbf{x}')\right] = \mathbb{E}_{f\sim\mathrm{GP}}\left[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{x}') - m(\mathbf{x}'))\right] = k(\mathbf{x}, \mathbf{x}')$$

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covariance

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- Hence, $f(\mathbf{x})$ and $f(\mathbf{x}')$ are jointly Gaussian distributed with **covariance** $k(\mathbf{x}, \mathbf{x}')$

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- Then, for any finite data set $x_1, \dots, x_N$ we have:

$$p(\mathbf{f}) = \mathcal{N}(\mathbf{f} \mid \mathbf{0}, \mathbf{K}_{ff})$$

Gaussian process regression

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- Let $\mathbf{f} = [f(x_1), \ldots, f(x_N)]$ we can write:

$$p(\mathbf{y} \mid \mathbf{f}) = \mathcal{N}(\mathbf{y} \mid \mathbf{f}, \sigma^2 \mathbf{I})$$

Gaussian process regression

- **Prior** on function f: $p(\mathbf{f}) = \mathcal{N}(\mathbf{f} \mid \mathbf{0}, \mathbf{K}_{ff})$
- Joint for **new observation** $x_\star$: $p(\mathbf{f}, f_\star) = \mathcal{N}\left(\begin{bmatrix}\mathbf{f} \\ f_\star\end{bmatrix} \mid \mathbf{0}, \begin{bmatrix}\mathbf{K}_{ff} & \mathbf{K}_{ff_\star} \\ \mathbf{K}_{f_\star f} & K_{f_\star f_\star}\end{bmatrix}\right)$

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What if the data has noise?

Gaussian process regression - putting everything together

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- **Likelihood**: $p(\mathbf{y} \mid \mathbf{f}) = \mathcal{N}(\mathbf{y} \mid \mathbf{f}, \sigma^2\mathbf{I})$ (Noise model)

Gaussian process regression - putting everything together

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- **Likelihood**: $p(\mathbf{y} \mid \mathbf{f}) = \mathcal{N}(\mathbf{y} \mid \mathbf{f}, \sigma^2\mathbf{I})$
- Joint model: $p(\mathbf{y}, f_\star, \mathbf{f}) = p(\mathbf{y} \mid \mathbf{f})\, p(f_\star, \mathbf{f})$

How do we make predictions?

Gaussian process regression - putting everything together

- To make predictions, we want to compute $p(f_\star \mid \mathbf{y})$ and marginalise out (remove) the dependence on $\mathbf{f}$.

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- **Predictions** with Gaussian process regression is again conditioning:

$$p(f_\star \mid \mathbf{y}) = \mathcal{N}\left(f_\star \mid \mathbf{K}_{f\star f}(\mathbf{K}_{ff} + \sigma^2 \mathbf{I})^{-1} [\mathbf{y}], K_{f\star f\star} - \mathbf{K}_{f\star f}(\mathbf{K}_{ff} + \sigma^2 \mathbf{I})^{-1} \mathbf{K}_{f\star f}^T \right)$$

Gaussian process regression - putting everything together

- Predictions **without** noise model:

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Implementing Gaussian process regression

Gaussian process regression

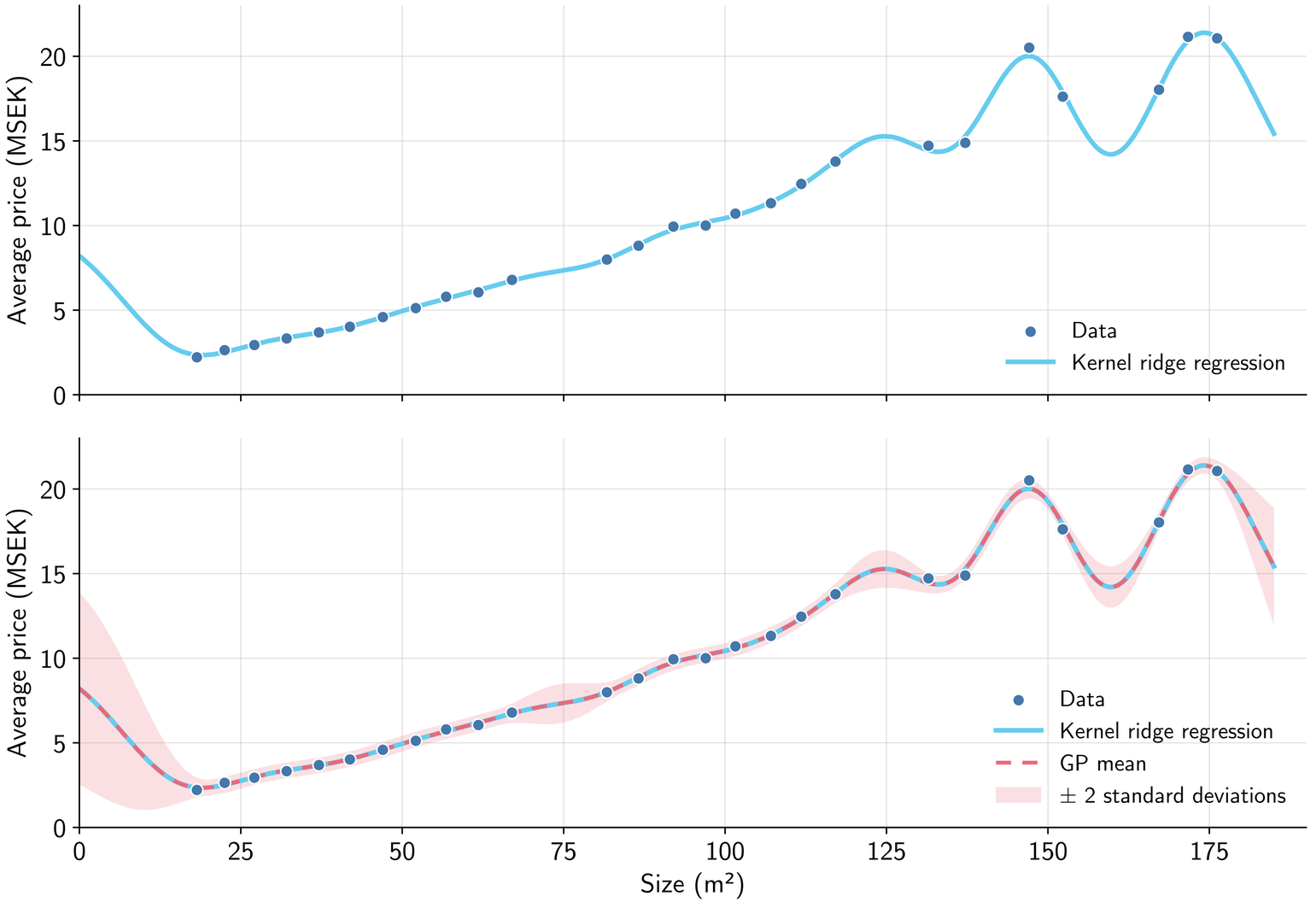
```python
# reuse rbf_kernel and gram_matrix from the last lecture
sigma2 = 0.1                              # noise variance σ²

K = gram_matrix(X, X)                     # K_ff
K_s = gram_matrix(X_star, X)              # K_f*f
K_ss = gram_matrix(X_star, X_star)        # K_f*f*
A = K + sigma2 * np.eye(len(X))           # K_ff + σ²I

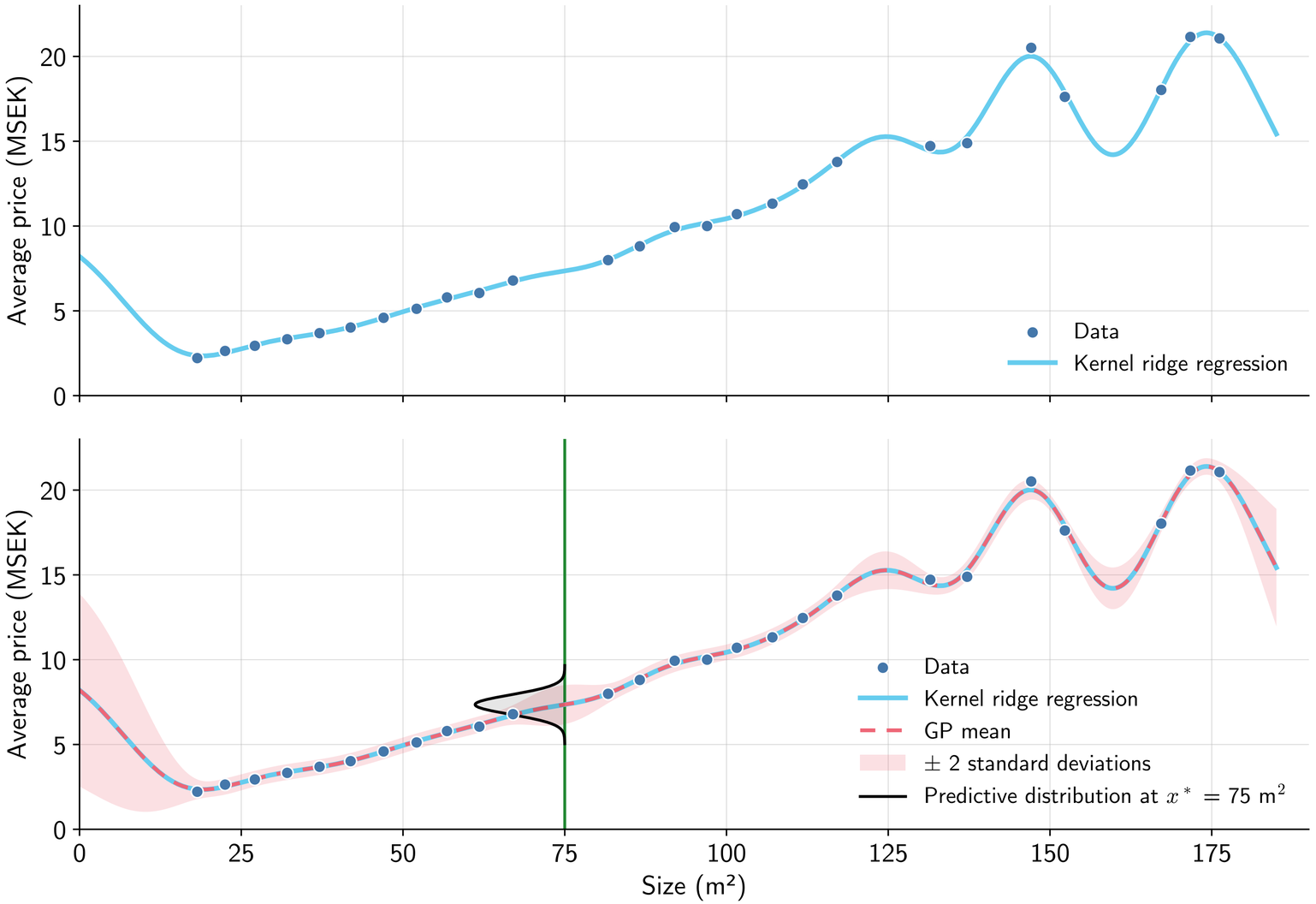
mean = K_s @ np.linalg.solve(A, y)
cov = K_ss - K_s @ np.linalg.solve(A, K_s.T)
std = np.sqrt(np.diag(cov))
```

https://github.com/traml-lab/id2214-lectures/blob/main/lecture6-gaussian-process.py

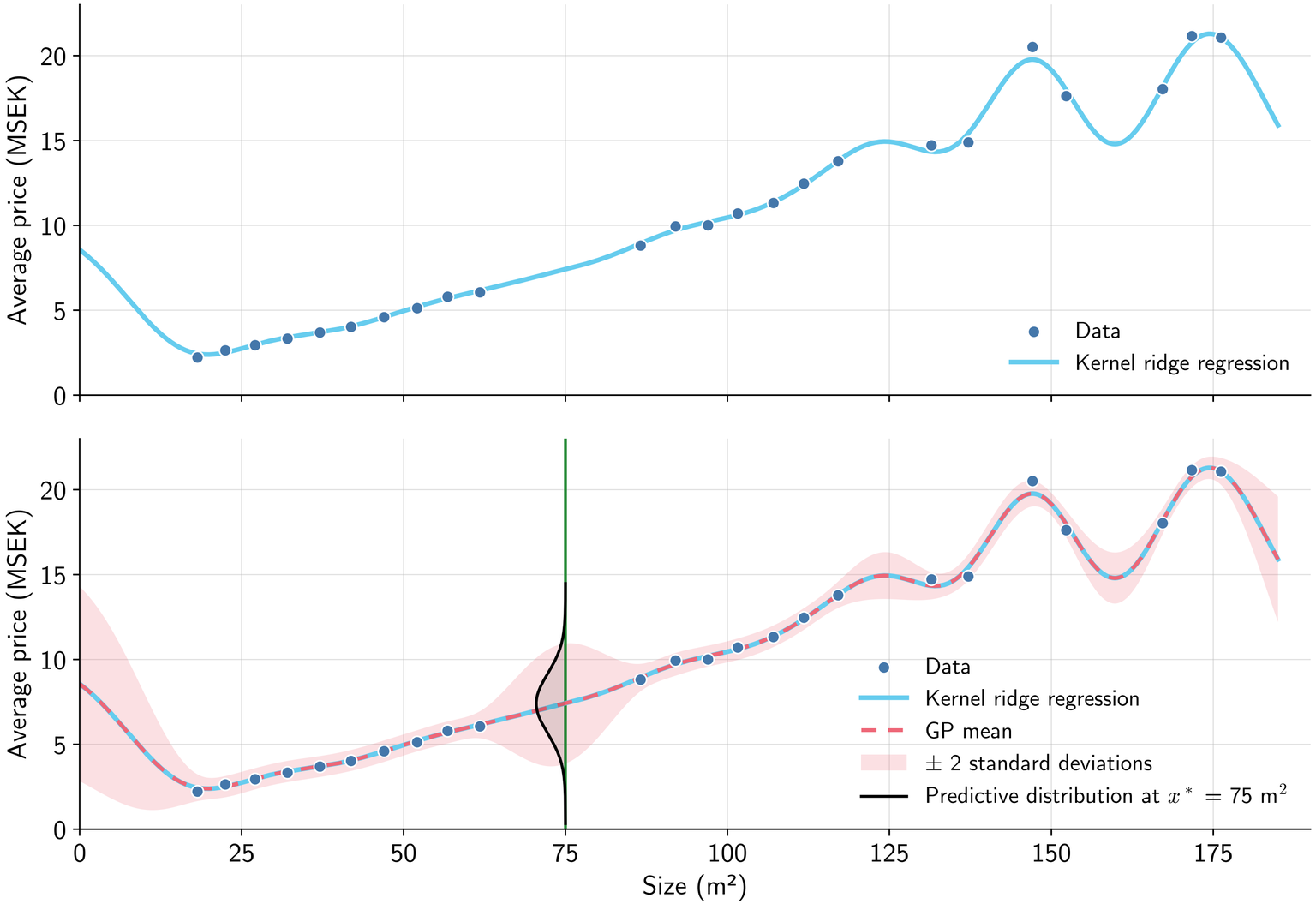
Gaussian process regression



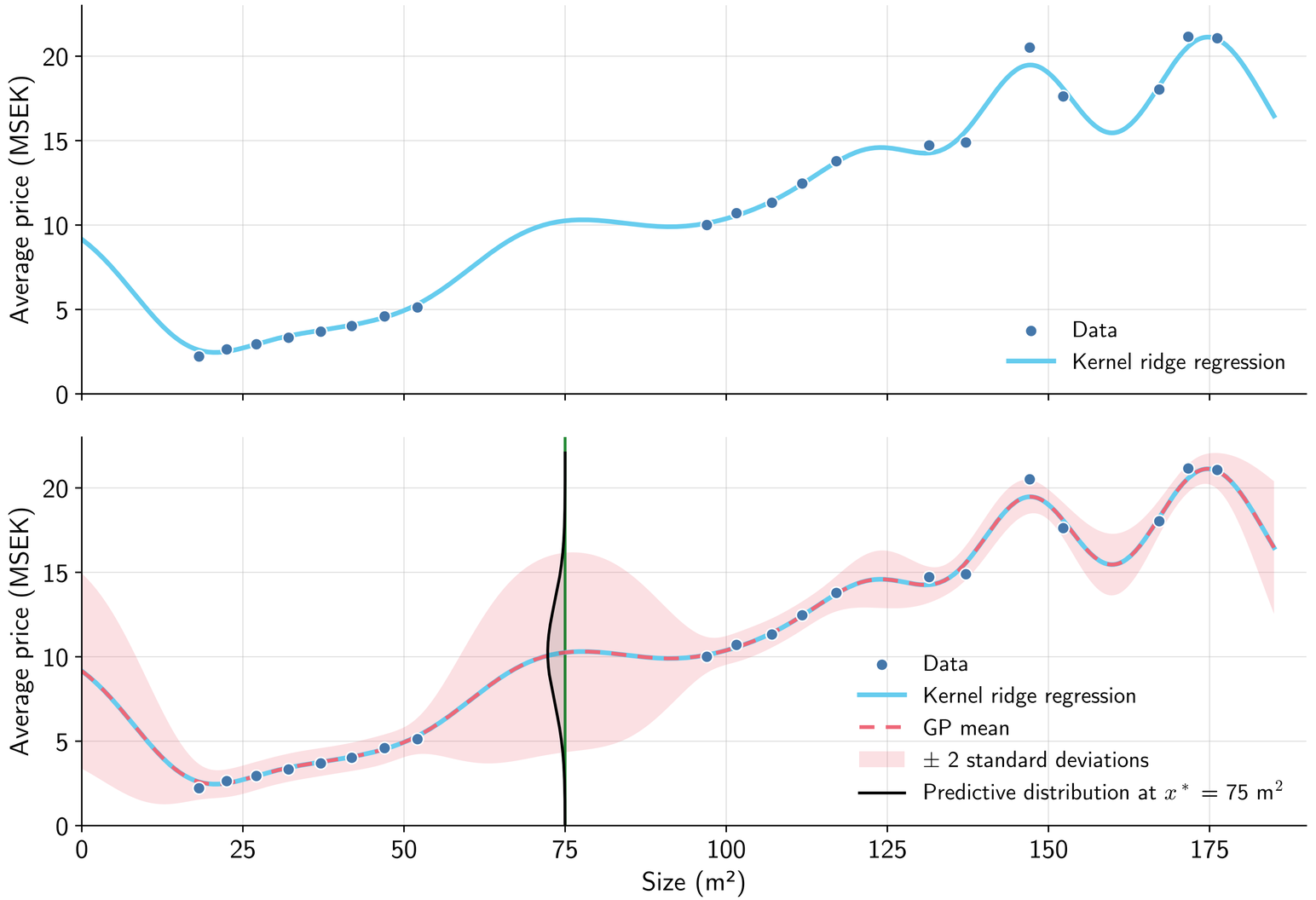
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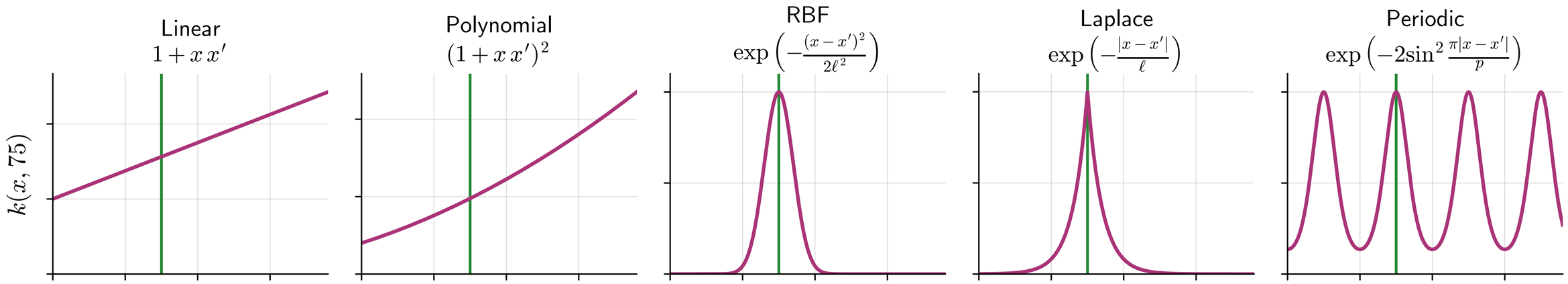
Gaussian process regression



Gaussian process regression



Kernel functions



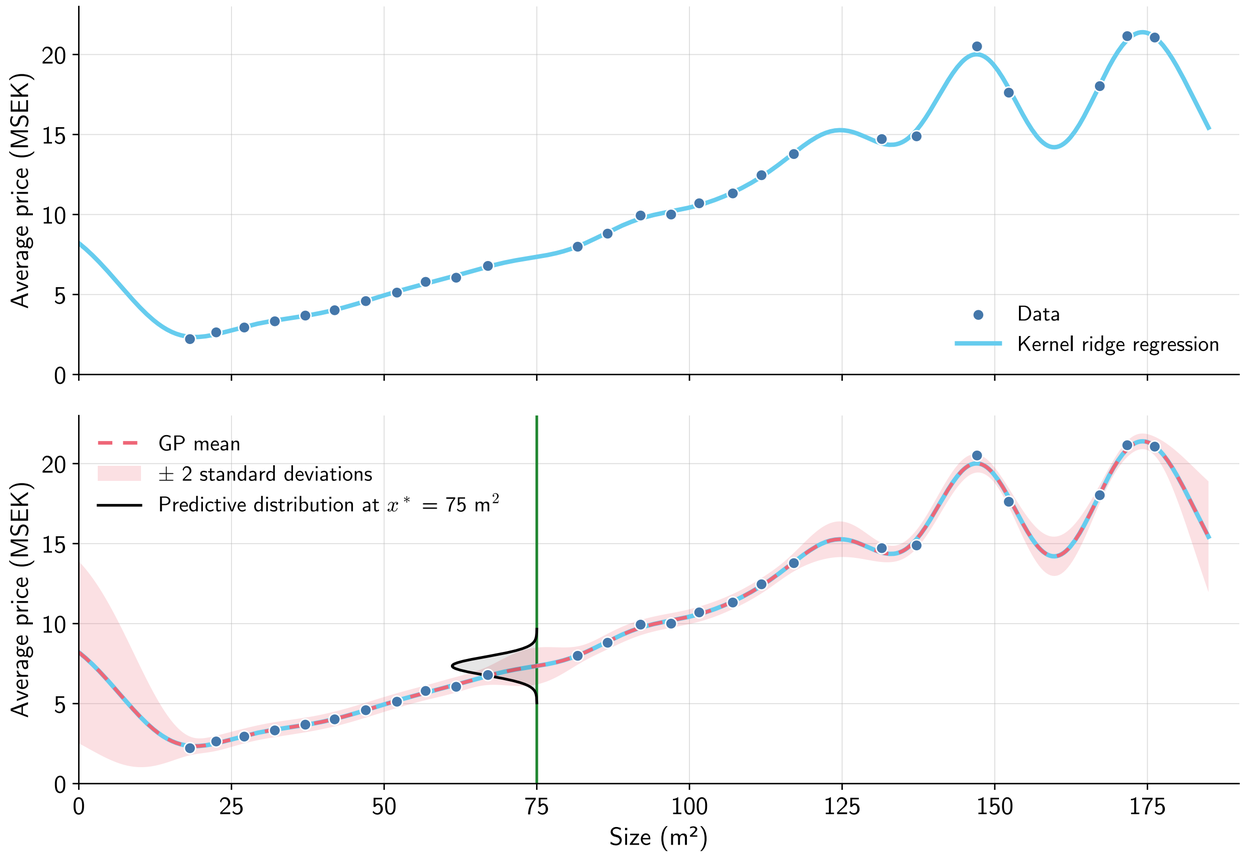
- In Gaussian processes, the kernel function is also a design choice!

$$\mathbf{K}_{ff} = \begin{bmatrix} k(x_1, x_1) & \dots & k(x_1, x_N) \\ \vdots & k(x_n, x_{n'}) & \vdots \\ k(x_N, x_1) & \dots & k(x_N, x_N) \end{bmatrix}$$

$$p(f_\star \mid \mathbf{y}) = \mathcal{N}\left(f_\star \mid \mathbf{K}_{f\star f}(\mathbf{K}_{ff} + \sigma^2\mathbf{I})^{-1}\left[\mathbf{y}\right], K_{f\star f\star} - \mathbf{K}_{f\star f}(\mathbf{K}_{ff} + \sigma^2\mathbf{I})^{-1}\mathbf{K}_{f\star f}^T\right)$$

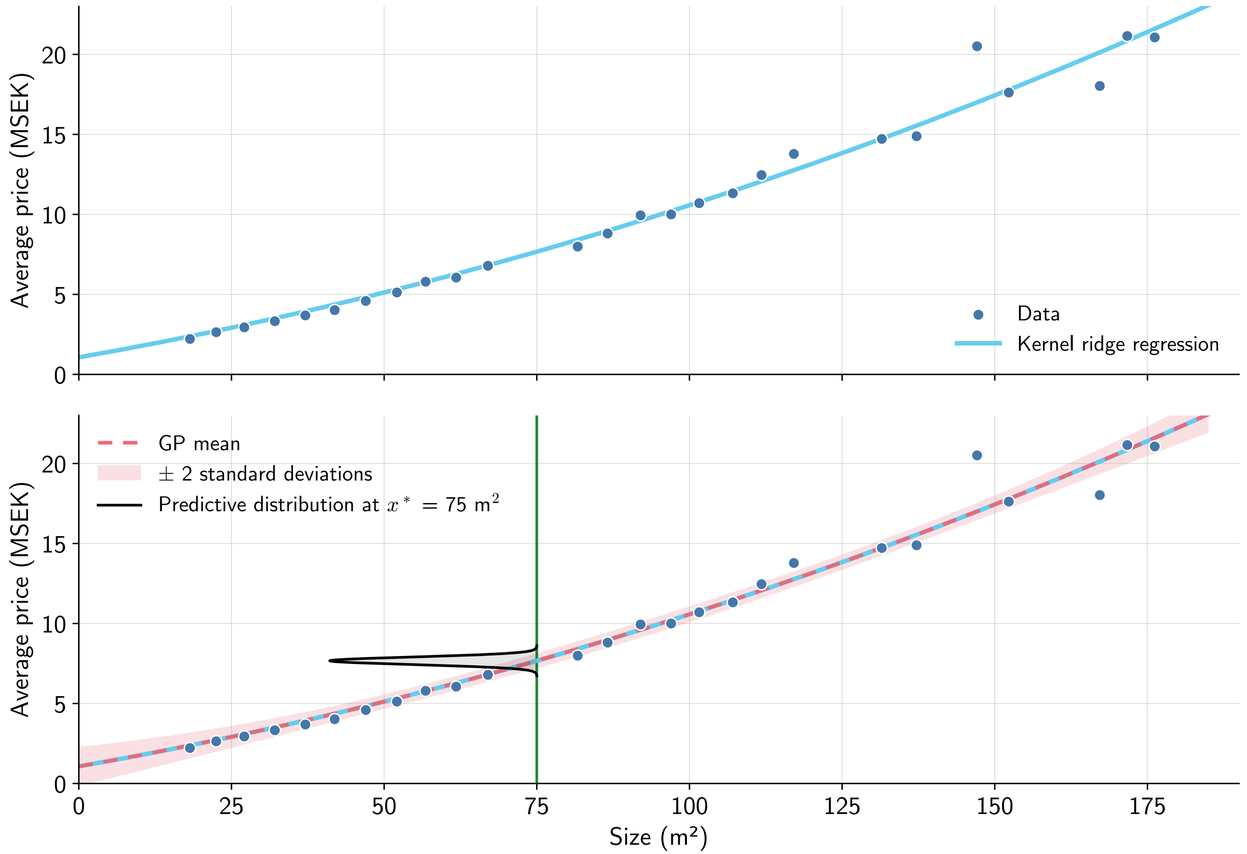
Gaussian process regression - kernel functions

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\ell^2}\right)$$



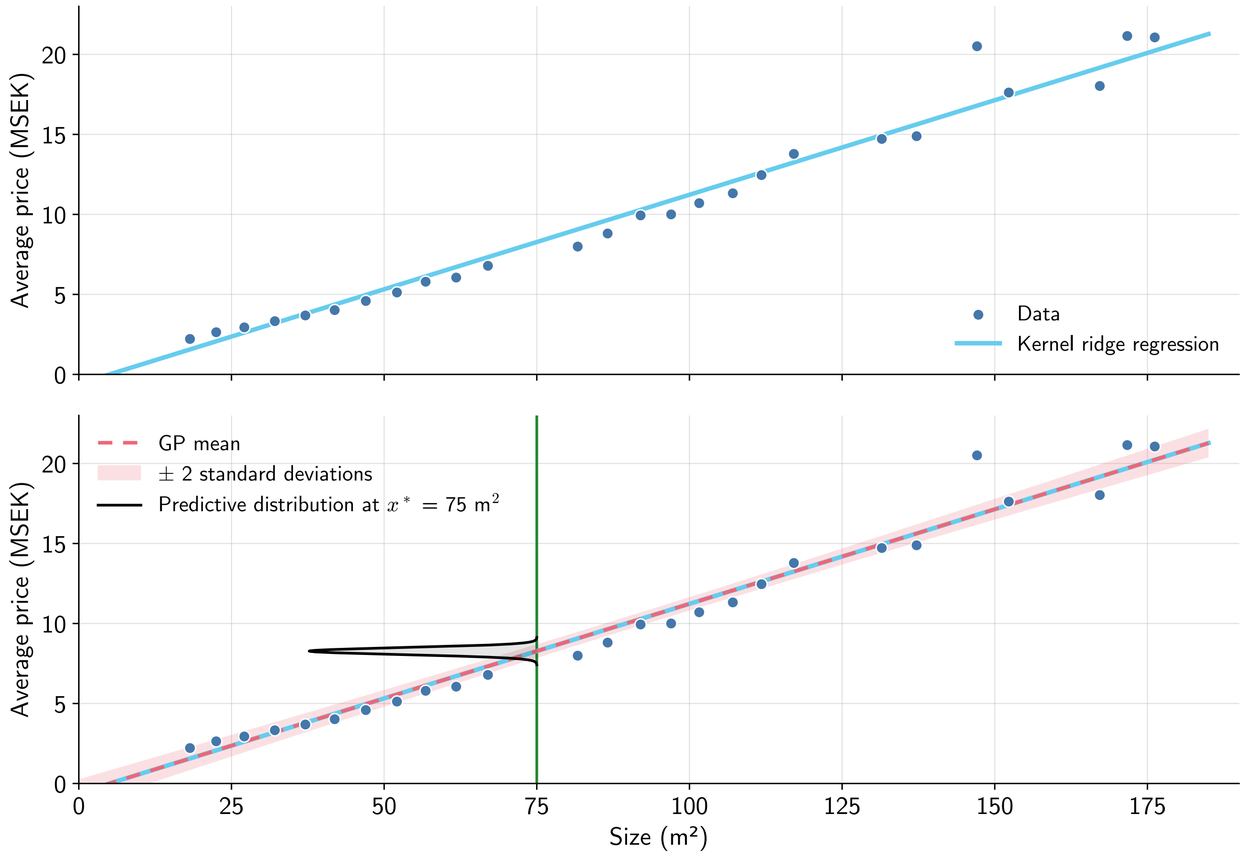
Gaussian process regression - kernel functions

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \left(1 + \mathbf{x}^T \mathbf{x}'\right)^2$$



Gaussian process regression - kernel functions

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- A GP is fully specified by its mean $m(\,\cdot\,)$ and its covariance/kernel function $k(\,\cdot\,,\cdot\,)$, but we typically assume a zero-mean $m(\mathbf{x}) = \mathbf{0}$ function for simplicity

- Predictions are conditional probabilities of multivariate Gaussians:

$$p(f_\star \mid \mathbf{y}) = \mathcal{N}\left(f_\star \mid \mathbf{K}_{f\star f}(\mathbf{K}_{ff} + \sigma^2\mathbf{I})^{-1}\big[\mathbf{y}\big], K_{f\star f\star} - \mathbf{K}_{f\star f}(\mathbf{K}_{ff} + \sigma^2\mathbf{I})^{-1}\mathbf{K}_{f\star f}^T\right)$$

The End

